



D.3 Motion in electromagnetic fields

Fields · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A charged particle moving perpendicular to a uniform magnetic field experiences a magnetic force of constant magnitude always perpendicular to its velocity, which provides centripetal force and causes circular motion: $qvB = mv^2/r$, giving orbital radius $r = mv/(qB)$, independent of the angle traversed (period $T = 2\pi m/(qB)$ is independent of speed). In combined electric and magnetic fields (as in a velocity selector), a charged particle travels in a straight line undeflected when the electric force balances the magnetic force, $qE = qvB$, giving selected speed $v = E/B$. In a uniform electric field alone (as between charged plates), a charged particle undergoes projectile-like motion: constant velocity along the field-free direction and constant acceleration $a = qE/m$ along the field direction.

Worked example: follow the reasoning

A proton ($q = 1.6 \times 10^{-19}$ C, $m = 1.67 \times 10^{-27}$ kg) moves at 2.0×10^6 m·s⁻¹ perpendicular to a magnetic field of 0.50 T. Find the radius of its circular path.

$$r = mv/(qB) = (1.67 \times 10^{-27})(2.0 \times 10^6)/[(1.6 \times 10^{-19})(0.50)]$$

$$= (3.34 \times 10^{-21})/(8.0 \times 10^{-20}) = 0.0418 \text{ m.}$$

Answer: $r \approx 4.2 \times 10^{-2}$ m.

Quick check · multiple choice

In a velocity selector, a charged particle travels undeflected through crossed electric and magnetic fields. What is the relationship between the electric field E , magnetic field B , and particle speed v for this to happen?

A $E = vB$

B $E = v/B$

C $E = B/v$

D $E = v^2B$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An electron ($q = 1.6 \times 10^{-19}$ C, $m = 9.11 \times 10^{-31}$ kg) enters a uniform magnetic field of 0.020 T, moving perpendicular to the field at a speed of 3.0×10^6 m·s⁻¹. [8]
- (a) Calculate the magnetic force on the electron as it enters the field.
- (b) Calculate the radius of the circular path followed by the electron.
- (c) Calculate the period of the electron's circular motion, and explain why this period is independent of the electron's speed.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $F = qvB = (1.6 \times 10^{-19})(3.0 \times 10^6)(0.020) = 9.6 \times 10^{-15} \text{ N}$.

(b) $r = mv/(qB) = (9.11 \times 10^{-31})(3.0 \times 10^6)/[(1.6 \times 10^{-19})(0.020)] = (2.733 \times 10^{-24})/(3.2 \times 10^{-21}) = 8.54 \times 10^{-4} \text{ m}$.

(c) $T = 2\pi m/(qB) = 2\pi(9.11 \times 10^{-31})/(3.2 \times 10^{-21}) = 5.723 \times 10^{-30}/3.2 \times 10^{-21} = 1.79 \times 10^{-9} \text{ s}$. This is independent of speed because although a faster electron has a larger radius, it also covers a proportionally longer circumference in the same time, since both r and the circumference $2\pi r$ scale linearly with v , leaving T unchanged.

Quick check answer

A. Undelected motion requires the electric force to balance the magnetic force: $qE = qvB \rightarrow E = vB$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.